

Zikki Qing

+61434564980 qingzqi6@gmail.com

<https://www.linkedin.com/in/zikki-qing-191186330/>

personal website: <https://portfolio-five-black-0rx8xlcgf1.vercel.app/>

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C#

Frameworks & Web: React, Next.js, React Native, Node.js, Django, .NET, ASP.NET Core

Databases: PostgreSQL, Redis

Cloud & DevOps: AWS, Azure, Linux, Docker, GitHub Actions, CI/CD

Testing & Quality: Cypress, Jest, pytest, K6, Automated Testing

AI / Data: Prompt Engineering, LLM Workflow Validation, AI Workflow Evaluation, Performance Benchmarking

Systems: Distributed Systems, Client-Server Architecture, TCP Communication, Fault Tolerance, State Synchronisation

PROFESSIONAL EXPERIENCE

Marketing Simplified - AI Marketing Workflow Platform(<https://zmarkio.com/>)

Mar 2026 - Present

Full-Stack Developer

Accomplishments:

- Designed a PostgreSQL audit logging system with composite indexes and database-level immutability controls, improving query performance and ensuring reliable tracking of workflow actions.
- Built an AI analysis context feature using Redis TTL storage, allowing users to guide LLM output direction while keeping short-lived request context ephemeral and avoiding unnecessary PostgreSQL writes.
- Built customer-facing React components in Next.js 14 App Router, including a filterable audit log timeline and meeting workflow UI with real-time validation feedback.
- Designed and implemented a human-in-the-loop anomaly review workflow, replacing automatic AI-generated decision creation with user-confirmed approval to reduce false-positive decisions and improve reliability of downstream workflow automation.

RhombusAI

Nov 2025 - Feb 2026

Full Stack Software Engineer Intern

Accomplishments:

- Improved reliability of an AI-driven data platform by diagnosing inconsistent LLM-generated workflows across React frontend, backend services, and REST APIs, reducing workflow failure rate by 15-25% and improving stability of user-facing outputs.
- Refined prompt behaviour and implemented validation checks for LLM-generated workflows, improving output consistency and reducing recurring execution errors across multiple data scenarios.
- Built Cypress smoke tests and nightly regression tests in GitHub Actions, reducing recurring frontend and workflow bugs by 30-40% and increasing test coverage of core workflows.
- Conducted API-level benchmarking and workflow validation to identify performance bottlenecks, improving pipeline completion time and overall system responsiveness.
- Collaborated with QA and cross-functional stakeholders to turn recurring workflow issues into clear test cases, validation rules, and regression coverage.

SELECTED PROJECTS

Kulaa – Cross-Platform Food Discovery & Booking Platform

- Built a cross-platform mobile and web application for Kulaa using React Native and Next.js, delivering core user-facing systems including search, booking, and profile management within an 8-week development cycle.
- Implemented REST API integrations and asynchronous frontend data flows, ensuring consistent user experiences across mobile and web interfaces under varying data and network conditions.
- Translated Figma designs into reusable frontend components and worked with stakeholders to refine MVP requirements, enabling faster iteration and reducing rework during development.

Distributed Multiplayer System

- Designed a custom TCP messaging protocol with acknowledgements, retry handling and role-based permissions, preventing duplicate updates and maintaining a consistent shared state across 10+ concurrent clients.
- Built a multi-threaded client–server architecture with dedicated worker threads for connection handling and state broadcasting, supporting concurrent users and synchronising newly joined clients within a single refresh cycle.
- Implemented fault-tolerant recovery logic for disconnects and reconnections, improving system reliability and reducing manual session recovery during multiplayer use.

EDUCATION

University of Melbourne

Master of Information Technology

2024.7 - 2026.6